

Industry Intelligence Agent

Interview Preparation Guide

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This guide summarises project architecture, KRI methodology, real news data with key numbers, the KRI pivot table, interview Q&A, and Deloitte JD mapping. **Bold numbers are directly quotable in interviews.**

1. What Is This Project?

A Python-based Industry Intelligence Agent that reads news RSS feeds, annual report PDFs, and company registry data, extracts Key Risk Indicators (KRI), generates a bilingual consulting-style Industry Intelligence Brief, and visualises results in a Streamlit dashboard.

Dimension	Details
Target audience	Consulting/strategy teams needing rapid industry risk overview
Primary output	KRI Evidence Table + Industry Intelligence Brief (ZH/EN)
Demo scenario	Taiwan semiconductor/AI server ecosystem (TSMC, MediaTek, Quanta...)
Tech stack	Python, pandas, Streamlit, Altair, pdfplumber, feedparser, reportlab
Code size	~1,200 lines (src/ + app.py)

One-sentence pitch

"I built a tool that turns raw news and annual reports into structured risk evidence and a bilingual consulting brief
- the kind of first-cut analysis a junior consultant would spend days on manually."

2. System Architecture & Data Flow

Step	Module	What It Does
1. Input	app.py sidebar	Company name, ticker, industry, PDF upload
2. Company Registry	src/company_registry.py	Lookup profile from CSV (name, ticker, industry)
3. News Collections	src/news_collector.py	Fetch RSS feeds, filter by keyword, score relevance
4. PDF Parsing	src/annual_report_reader.py	Extract text chunks from annual report PDF
5. KRI Extraction	src/kri_extractor.py	Keyword match -> 12 risk categories + severity score
6. Trend Notes	src/industry_trend_reader.py	Convert combined text to structured trend JSON
7. Report	src/report_generator.py	Generate ZH/EN Markdown brief from all evidence
8. Dashboard	app.py	Streamlit: metrics, Altair charts, pivot table, download

Key Design Principle: Guardrails

- Facts from sources - company registry, news summaries, annual report sentences
- Analytical interpretation - risk prioritisation, business implications
- Recommended follow-up - client interview questions and data validation

Every output is labelled by type. This prevents hallucination and mirrors consulting report standards.

3. KRI Extraction Methodology

KRI vs KPI

KPI measures current performance (e.g. TSMC revenue +45.2% YoY). **KRI** is an early warning signal of risk before it hits financials (e.g. 'Helium prices doubled' = supply chain risk). This system extracts KRI evidence from text using 12-category keyword dictionaries.

12 KRI Categories

Category	Key Trigger Words	Demo Evidence
supply chain risk	shortage, logistics, supplier, delivery delay	Suez Canal / Hormuz closed; helium 2x price
geopolitical risk	tariff, export controls, cross-strait, war	US 50% tariff China chips; Taiwan 15%
customer concentration risk	major customer, top 5 customers, key customer	Key customer alert: NVIDIA & Broadcom
ESG / sustainability risk	energy usage, carbon, emissions, renewable	92% energy imported; 11-day LNG reserve
regulatory risk	regulation, compliance, sanction, fine	China: Ga, Ge, graphite export controls
profitability risk	gross margin, pricing pressure, cost increase	Arizona fab +30% cost vs Taiwan
inventory risk	inventory, obsolete, stock level, turn	Turnover uncertainty -> order timing shifts
cash flow risk	cash flow, capex, free cash flow	USD 165B+ US investment commitments
liquidity risk	liquidity, working capital, current ratio	Volatile orders -> WC pressure
receivables risk	accounts receivable, bad debt, credit loss	Enterprise payment terms vary
cyber / digital risk	cyber, data breach, ransomware, OT	Connected factories -> OT attack surface
leverage risk	debt, borrowings, interest expense, leverage	Large capex financed via debt markets

Severity Scoring

Level	Trigger Words	Score
HIGH (3)	"significant", "material", "severe", "critical", "sharp decline", "liquidity pressure"	3
MEDIUM (2)	"decline", "shortage", "delay", "loss", "uncertainty", "pressure", "disruption"	2
LOW (1)	Any KRI keyword without HIGH or MEDIUM words	1

4. The KRI Pivot Table - COUNTIFS Equivalent

What it is

The KRI Category x Severity Pivot Table shows how many evidence sentences belong to each risk category at each severity level. It is the single most important output for prioritising analyst review - equivalent to running Excel COUNTIFS across 36 combinations at once.

Excel equivalent: =COUNTIFS(kri_category, "supply chain risk", severity, "high")

Demo Output (8 KRI rows from real news)

KRI Category	High	Medium	Low	Total	Priority
supply chain risk	1	0	0	1	1st - validate immediately
geopolitical risk	1	0	0	1	1st - validate immediately
customer concentration risk	1	0	0	1	1st - validate immediately
ESG / sustainability risk	1	0	0	1	1st - validate immediately
regulatory risk	0	1	0	1	2nd - monitor closely
profitability risk	0	1	0	1	2nd - monitor closely
inventory risk	0	1	0	1	2nd - monitor closely
cyber / digital risk	0	0	1	1	3rd - review in next cycle

How to explain it in an interview

- HIGH severity -> validate first with management interview + financial data
- MEDIUM severity -> dashboard monitoring candidates
- Appears in news AND annual report -> management has already acknowledged the risk
- Appears only in news (not annual report) -> potential management cognition gap
- High total across categories -> systemic risk environment, not isolated events

5. Key Numbers to Quote in Interviews

All sourced from real news (Jan-May 2026). Cite the source when you quote.

Fact	Number	Source
TSMC March 2026 revenue YoY	+45.2%	HeyGoTrade, Apr 2026
TSMC March 2026 revenue	NTD 415.2B	HeyGoTrade, Apr 2026
TSMC dividend increase	+28%	247 Wall St, Feb 2026
TSMC global advanced node market share	>65%	Yahoo Finance, Apr 2026
Taiwan energy import dependency	97%	Yahoo Finance, Apr 2026
Taiwan LNG reserve (no imports)	11 days	Yahoo Finance, Apr 2026
Strait of Hormuz closure date	March 4, 2026	HeyGoTrade, Apr 2026
Helium price increase	2x (doubled)	HeyGoTrade, Apr 2026
US tariff on Chinese semiconductor imports	50%	Tom's Hardware, Apr 2026
Taiwan reciprocal tariff rate (US deal)	<=15%	CNBC, Jan 2026
Taiwan US investment commitment	USD 250B	CNBC, Jan 2026
TSMC Arizona fab cost premium vs Taiwan	+30%	247 Wall St, Feb 2026
TSMC Arizona total investment target	USD 465B / 11 fabs	abhs.in, Apr 2026
New China chip tariffs effective date	June 2027	Tom's Hardware, Apr 2026
US-China total tariff on Chinese goods	30%	Tax Foundation, May 2026

6. Deloitte JD Mapping

JD Requirement (ZH)	How This Project Demonstrates It
■■■■■■■■■■■■■■■	KRI evidence table, industry trend notes, and stacked pivot chart from real news data
■■■■■■■■■	Python pipeline: pandas wrangling, keyword NLP, severity scoring, Altair chart generation (~1,200 lines)
■■■■■■■■ AI ■■	Digital Transformation Opportunity Map with 5 concrete use cases (demand planning, supplier risk, WIP management, etc.)
■■■■■■■■ business insight	Bilingual ZH/EN brief separating facts / interpretation / recommended follow-up (consulting standard)
■■■■■	Can demo the Streamlit app, explain KRI methodology, and quote key numbers in 2-3 minutes

7. Interview Q&A; - Talking Points

Q: Tell me about your project.

An automated Industry Intelligence Agent. Given a company name and PDF annual report, it extracts 12 types of KRI from news and text, scores them by severity, generates a bilingual consulting brief, and visualises in an interactive dashboard. Demo focuses on TSMC and the Taiwan semiconductor/AI server ecosystem using real 2026 news.

Q: What is KRI and how is it different from KPI?

KPI measures current performance - TSMC revenue was +45.2% YoY in March 2026. KRI is an early warning signal. 'Helium prices doubled' is a KRI for supply chain risk. My system extracts KRI from text using 12 keyword categories and scores severity as High/Medium/Low based on word strength.

Q: How accurate is your KRI extraction?

Rule-based keyword matching - transparent, not a black box. For the demo, all 8 KRI rows were verified against real news (CNBC, Tom's Hardware, Yahoo Finance). In production, an LLM layer would improve recall on ambiguous sentences.

Q: Why Streamlit, not Power BI?

Streamlit lets me wire Python analysis directly to the UI. For a client deliverable, Power BI would be appropriate - I built dashboard-ready CSV export precisely for that handoff.

Q: What is the KRI Pivot Table?

It is a COUNTIFS-equivalent summary: for each of 12 risk categories, it shows how many evidence sentences are High/Medium/Low severity. Single most useful output for prioritising analyst review. Categories with HIGH count get validated first with management interviews. Categories only appearing in news (not annual report) suggest a management cognition gap.

Q: What would you improve with more time?

(1) LLM classifier to improve KRI recall on ambiguous text. (2) Financial data integration (TWSE API, Bloomberg) linking KRI signals to actual ratios. (3) Time-series view tracking how risk signals evolve across quarterly reports.

Q: How does this relate to Deloitte's work?

Deloitte advisory starts with rapid industry and company analysis before client interviews. This tool automates evidence-gathering and risk-categorisation, so consultants focus on validating findings, designing solutions, and building the business case. The KRI Evidence Table becomes a pre-interview hypothesis list.

8. How to Run

```
cd c:\Projects\Industry_intelligenceAgent\industry_intelligence_agent
streamlit run app.py
```

Opens at <http://localhost:8501>. Default loads TSMC demo data with real 2026 news. To analyse Apple: uncheck 'Use demo data', change Company to Apple, upload 10-K PDF.

Active File Structure (after cleanup)

Path	Purpose
app.py	Streamlit UI - main entry point
src/kri_extractor.py	KRI keyword extraction + severity scoring
src/report_generator.py	ZH/EN Markdown brief generation
src/news_collector.py	RSS feed fetching + keyword filter
src/annual_report_reader.py	PDF text extraction and chunking
src/summarizer.py	LLM or rule-based company brief generation
data/demo/	Pre-built demo data (real news, KRI, trends)
data/report/	Generated PDF (this file)
configs/rss_sources.yaml	RSS feed URLs - editable without code change